

KIRAN VIJAYKUMAR CHOUDHARI

AI / ML Engineer

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SUMMARY

AI / ML Engineer working at the intersection of GenAI and production infrastructure. Built multi-agent systems for autonomous incident analysis, LLM fallback architectures, and reliable deployments across cloud environments.

EXPERIENCE

AI Intern

Crafttech 360

Sep '25 — Nov '25
Bengaluru, India

Project: [Cheeko AI Toy](#) – An AI-powered screen free toy for kids

- Designed and deployed a **RAG pipeline** over structured textbook content, enabling **context-aware LLM responses** through embedding-based semantic retrieval.
- Designed and implemented **CI/CD pipelines** using **CircleCI** and **Azure** to automate ML and backend deployments, reducing manual intervention and improving deployment consistency.
- Deployed backend services on Azure using **PostgreSQL** for persistence and **Redis** for low-latency caching, improving response times for frequent queries.
- Containerized ML pipelines and backend services using **Docker** to standardize runtime environments and enable reliable cloud deployments.
- Monitored system behavior post-deployment and iterated on retrieval and inference components to **improve response consistency**.

PROJECTS

DokGuru - AI-Powered Multilingual Document Assistant [Link](#)

Nov '25 — Present

- Architected **production-grade RAG system** with **Flask**, **React**, and **PostgreSQL (pgvector)**, implementing intelligent caching and connection pooling to reduce API response times from multi-second to **sub-second latency**.
- Designed **document ingestion pipeline** with **1000-word chunking** and **200-word overlap**, using **parallel processing (4 workers)** to extract text, tables, and images from **multi-page PDFs** efficiently.
- Implemented **multilingual voice pipeline** supporting **3 languages** with **cascading TTS engines (Piper, EdgeTTS, gTTS)**, achieving **sub-second synthesis** compared to a 2+ second baseline.
- Engineered **multi-provider LLM inference layer** with **automatic fallback routing** and **connection pooling (max 20 concurrent)** to maintain high availability during provider outages and latency spikes.
- Deployed **zero-downtime CI/CD pipeline** on **Google Cloud Run** with **canary deployments**, automated health checks, and **JWT authentication** with **rate limiting (100 req/hour)**.

OpsPilot - AI-Powered Production Incident Analysis System [Link](#)

Dec '25 — Present

- Architected a **multi-agent AI system** with **Planner, Verifier, Fixer, and Remediation agents** to perform autonomous root cause analysis of production incidents, reducing diagnosis time from **hours to minutes**.
- Engineered a **multi-provider LLM inference system** with **automatic fallback routing** across **Ollama, OpenRouter and Gemini** ensuring **99.9% uptime** during provider outages and API rate limits.
- Built an **intelligent error pattern detection engine** analyzing **HTTP errors, exceptions, database failures, and timeouts**, with automated **severity classification (P0–P3)** based on frequency and impact.
- Implemented **multi-source production log ingestion** supporting **S3, Kubernetes, CloudWatch, and HTTP endpoints**, using **parallel processing** for rapid data collection during critical incidents.
- Designed **deployment correlation analysis** by parsing **Git history** to identify code changes within configurable time windows and correlate deployments with production errors.

SKILLS

AI / ML RAG pipelines, LLMs, Embeddings, Semantic Search, CNNs, Multi-Agent Systems, NLP

Programming Languages Python, JavaScript

Database & Caching SQL, MongoDB, PostgreSQL, Qdrant, Redis

Deployment/DevOps Docker, Azure, Google Cloud (Cloud Run), Vercel, Github Actions

Concepts Object-Oriented Programming, API Design, Authentication & Authorization

EDUCATION

B.E in Artificial Intelligence & Machine Learning, MVJ College of Engineering (GPA: 7.4)

Dec '21 — Jun '25

PUC in PCMB, Tungal Independent PU College Vijayapura (Result: 90%)

Jun '19 — Jul '21

SSLC, M E S High School Chadchan (Result: 86%)

Jun '18 — Apr '19

CERTIFICATIONS

[Oracle Cloud Infrastructure 2024 Generative AI Certified Professional](#), Oracle

Jul '24